

Homework 1

Stat 215A, Fall 2026

Due: Submit a `homework1.pdf` file to **Gradescope** by **Saturday, September 19 23:59**

1 Hypothesis testing, the t -distribution

Imagine we observe $(x_1, y_1), \dots, (x_n, y_n)$ where (x_i, y_i) are multivariate normal with mean (μ_x, μ_y) , $\text{Var}(x_i) = \text{Var}(y_i) = \sigma^2$ and correlation ρ . We are interested in testing the null hypothesis that $\mu_x = \mu_y$.

Under the null hypothesis we know

$$t = \frac{(\bar{x} - \bar{y})}{s_{pooled} \sqrt{2/n}}$$

is distributed as a Student's t with $2n - 2$ degrees of freedom, where s_{pooled} is the pooled sample standard deviation. See any undergraduate text (or Wikipedia page "Student's t-test") if you are unfamiliar with the t distribution.

1. Write s_{pooled} in terms of $x_i, y_i, \bar{x}$ and $\bar{y}$ (this is a standard definition)
2. What is the expectation of s_{pooled}^2 ?
3. The statement above (on the t -statistic) isn't quite right. Are any additional assumptions needed?

Consider doing a paired t -test with the same data. The test statistic here is

$$t_{paired} = \frac{(\bar{x} - \bar{y})}{s_{diff} \sqrt{1/n}}.$$

4. Write s_{diff} in terms of $x_i, y_i, \bar{x}$ and $\bar{y}$. (another standard definition)
5. What distribution does t_{paired} have?
6. What is the expectation of s_{diff}^2 ?
7. Compare $s_{diff}^2 \frac{1}{n}$ to $s_{pooled}^2 \frac{2}{n}$. When is $s_{diff}^2 \frac{1}{n} < s_{pooled}^2 \frac{2}{n}$? When is $E(s_{diff}^2 \frac{1}{n}) < E(s_{pooled}^2 \frac{2}{n})$?
8. From these computations, what do you learn?

2 Questions from Freedman

In Freedman, do questions 1 - 5 and 9 starting on page 13. This may sound like a lot of work. However, once you do the reading, each question should have a straight-forward answer.

Please look into the following map: <https://commons.wikimedia.org/wiki/File:Snow-cholera-map-1.jpg>. This is the map made by John Snow regarding the Broad Street pump. Each small block marks a cholera patient.

- How would you transform this display into numerical measures?
- What would be gained quantifying the effects? What would be lost?

3 Miscellaneous

- Contrast the data science life cycle outlined by the PCS framework to an unstructured data science process (this can be either abstractly or with a fixed data example). What is an application area where you believe the PCS framework can have impact?
- Give an example of a methodological problem that Tukey names as a significant question in data analysis and you believe is still an open problem. Explain where and when this problem arises in real data.
- What are some dangers of an open feedback loop between theory and practice in science/data science?